Active Robot Vision with Perception-Aware Navigation for Worker Monitoring

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ABSTRACT

Ensuring reliable object detection is essential for vision-based safety monitoring on construction sites. Fixed-camera systems often encounter challenges from occlusions caused by obstacles and workers' postures. While mobile robots can address these limitations through dynamic repositioning, the incorporation of their visual detection range into navigation strategies remains an open challenge, limiting detection consistency in complex environments. This study proposes an active monitoring navigation method based on a detection field, which enables robots to dynamically adjust their positions to keep targets within the detectable range of the vision sensor. Using deep reinforcement learning, the navigation policy optimizes the robot's movements around workers to maintain optimal visibility. This study also explores how different detection field configurations impact social distance in worker-robot interaction, while enhancing navigation and detection reliability. By optimizing the detection field, this approach advances active monitoring systems, promoting more reliable and safer worker-robot interactions in complex and dynamic construction environments.

INTRODUCTION

Advancements in computer vision technology have significantly enhanced the use of vision sensors for worker safety monitoring on construction sites, particularly for verifying Personal Protective Equipment(PPE) compliance through object detection models. In real-world applications, the confidence level of object detection is critical, as systems like autonomous robots rely on high-confidence detections to make informed decisions and take appropriate actions(Selvaraj et al., 2020). Improving detection confidence remains a primary challenge, as poor or unreliable confidence levels may lead to potential risks that compromise worker safety. However, fixed-viewpoint monitoring systems face significant challenges due to occlusion, where target objects are temporarily blocked or fall outside the camera's Field of View(FoV). These issues, often exacerbated by variations in angle, position, or environmental conditions, present

critical challenges to achieving accurate detection and classification in vision-based(Paneru & Jeelani, 2021). As fixed cameras are constrained by their installation locations and restricted FoV, these factors severely impact detection reliability and overall system performance.

To overcome these challenges, recent studies have explored an active monitoring approach utilizing UAVs or UGVs to dynamically adjust viewpoints. Research using UGVs (Cai et al., 2023) has proposed a path planning method that predicts worker movements and avoids collisions, while UAV-based systems have been employed to enhance worker detection accuracy in large-scale construction sites(Han et al., 2025). Studies on active perception (Tosidis et al., 2022) have proven that mobile agents utilizing Deep Reinforcement Learning(DRL)-based learning can significantly improve facial recognition accuracy. Our previous study(Park et al., 2024) has also demonstrated that enabling robots to navigate within specific angular ranges around workers can mitigate occlusion challenges and enhance object detection by providing diverse observation angles, even in obstacle-filled environments. However, it primarily focused on positioning robots to observe workers from various angles, with less emphasis on whether the targets were consistently within the robot's detection range or if its final position ensured reliable detection. These gaps in detection reliability raise uncertainties that compromise the monitoring effectiveness. To address this, it is necessary to resolve such uncertainties by incorporating the concept of the robot's field of view (FoV), which plays a critical role in effectively overcoming occlusion problems(Seits et al., 2022).

Another key consideration is maintaining appropriate proxemics during robot navigation(Samarakoon et al., 2022). Simply maximizing separation distances or stopping at safe positions may enhance physical safety, but can result in unreliable detection. Thus, balancing the distance needed for effective monitoring with the distance ensuring both physical and psychological worker safety is crucial. To explore this and better understand human-robot interactions, this study implemented variable distance configurations in reinforcement learning(RL) to analyze these dynamics.

Building on these considerations, this research aims to develop and evaluate an active monitoring navigation system utilizing the robot's field of view (FoV) to address occlusion challenges and improve detection reliability. The dynamic detection field introduced in this study enhances detection accuracy in occlusion scenarios by leveraging real-time perceptual information. Additionally, the adaptive proximity control mechanism integrated with the detection field optimally adjusts the robot's distance based on the occlusion status of the target, effectively improving worker safety. By achieving these contributions, this study advances active monitoring systems in construction sites and enhances reliable worker safety monitoring technologies.

BACKGROUND

Challenges of Computer Vision Applications for Safety Monitoring. In recent, surveillance cameras have become a standard tool for on-site worker monitoring, with deep learning-based methods effectively recognizing workers(Luo et al., 2020). However, one of the key challenges faced by these vision-based systems is their sensitivity to occlusion, which can affect object detection reliability and, in turn, impact worker safety(Nath et al., 2020). As a solution to this issue, researchers have explored strategies such as leveraging partial skeleton information to reconstruct occluded structures and employing data augmentation techniques(Chen et al., 2023; Li & Li, 2022). While these methods have shown potential, their applicability is often restricted to specific targets and environmental conditions, limiting their generalizability.

To fundamentally mitigate occlusion challenges, an active searching through robot mobility has emerged as a promising solution(Shang & Ryoo, 2024). Inspired by strategies for active exploration and information gathering, this approach enables robots to intelligently interact with their environment, actively gather information in uncertain situations, and enhance navigation decision-making reliability(Wang et al., 2020). Our previous research(Park et al., 2024) introduced a navigation method utilizing the angle between the robot and target objects to optimize observation positions by navigating around workers. While this approach effectively improved coverage, it primarily focused on the robot's position without accounting for obstacles within its detection range or verifying the presence of occluded objects. This could lead to inconsistencies in reliably detecting targets at the robot's final position. Thus, even if the target object is within the visible range, it may remain undetected due to factors such as poor model performance, obstructed views, or excessive distance. Therefore, reliable object detection in active monitoring requires integrating the detection range into the navigation policy design. By incorporating detection range considerations into decision-making, this approach reduces uncertainties and enhances the reliability of object detection in dynamic environments.

Reinforcement Learning-Based Navigation. Robotic navigation in complex and dynamic environments remains a significant challenge. Conventional goal-oriented navigation often relies on pre-stored maps and focuses on reaching predefined destinations, limiting their ability to proactively adapt to dynamic human activity(Xu & García de Soto, 2022). As a result, conventional methods are not well-suited for actively monitoring workers in occluded situations. Meanwhile, DRL has emerged as a powerful tool for adaptive navigation, enabling robots to interact with their surroundings and refine path-planning strategies through experiential learning(Taheri et al., 2024). This approach significantly enhances the stability and precision of robotic navigation, particularly in unpredictable environments(Arulkumaran et al., 2017). Reinforcement learning has been successfully applied to navigation tasks, enabling robots to optimize their behaviors and paths through reward-driven learning mechanisms(Pan et al., 2021). By continuously adapting to their surroundings, robots equipped with DRL-based navigation systems can move beyond goal-oriented path planning toward task-oriented adaptation, ensuring robust and efficient deployment in real-world, safety-critical workplaces. Therefore, this study proposes a DRL-based navigation strategy that enables robots to actively reposition themselves in occluded environments and secure optimal viewpoints. Through autonomous exploration and adaptive positioning, this approach enhances worker monitoring in complex construction sites.

Human-Robot Proxemics. As robots increasingly operate alongside human workers, ensuring safe and effective interactions has become a critical concern. The presence of robots introduces potential risks, including physical collisions and perceived threats, which must be mitigated to ensure both physical and psychological safety(Kim et al., 2020). In dealing with this challenge, it is essential to define optimal proxemics — the spatial boundaries that guide human-robot interactions. For some construction robot applications, such as progress monitoring and safety management, workers do not need to interact closely with the robot. These tasks are therefore classified as distant human-robot interactions(Sun et al., 2023). While short paths for distant interactions are generally preferred for operational efficiency, safety considerations take precedence in environments where simultaneous activities occur in confined spaces(Kim et al., 2024). To navigate these competing requirements, this study introduces the concept of the detection field, enabling robots to dynamically adjust their spatial relationship with workers.

METHODOLOGY

This study addresses the critical challenge of occlusion in worker monitoring, where PPE worn by workers becomes partially or fully obscured due to factors such as worker posture, physical obstacles, or limitations in sensor configurations. To address this issue, the problem is formulated as viewpoint optimization in a dynamic environment. The goal is to enable a robotic system to reposition itself dynamically, maintain consistent visibility of monitoring targets such as PPE compliance, and ensure safe navigation. The study employs reinforcement learning within a simulation environment to develop an active navigation system capable of adapting to dynamic construction settings. Central to this approach is the concept of a refined "detection field," defined as the subset of the robot's field of view where object detection models can reliably identify targets. Using this framework shown in Figure 1(a), the DRL-based system dynamically adjusts the robot's position to optimize its monitoring perspective. By proactively resolving occlusion through lateral repositioning and adaptive navigation, the system ensures that PPE compliance remains detectable, even under challenging and uncertain conditions.

Detection Field. The concept of the detection field is introduced to address limitations in previous research, particularly their inability to ensure reliable detection. The detection field is based on the specifications of the robot's vision sensors and the performance of its object detection model. A critical challenge lies in balancing detection reliability and proximity to workers, as closer distances can enhance detection accuracy but raise concerns about safety and comfort.

To explore this trade-off, detection field configurations were developed by varying the robot's detection range. Each configuration maintained a horizontal range of 60°, aligned with the camera's horizontal FoV specifications, while incrementally increasing the vertical range from 3m to 7m. These configurations, labeled v3 through v7, were designed to identify optimal detection ranges that ensure reliable detection while maintaining a safe distance for workers.

Environmental Setup. The learning environment was created using the Unity Engine. This simulation environment includes a worker represented as a human model, whose posture may obscure the visibility of detection targets such as PPE compliance. The detection targets are randomly positioned behind the worker, obscured from the front but detectable when the robot moves to the worker's left or right.

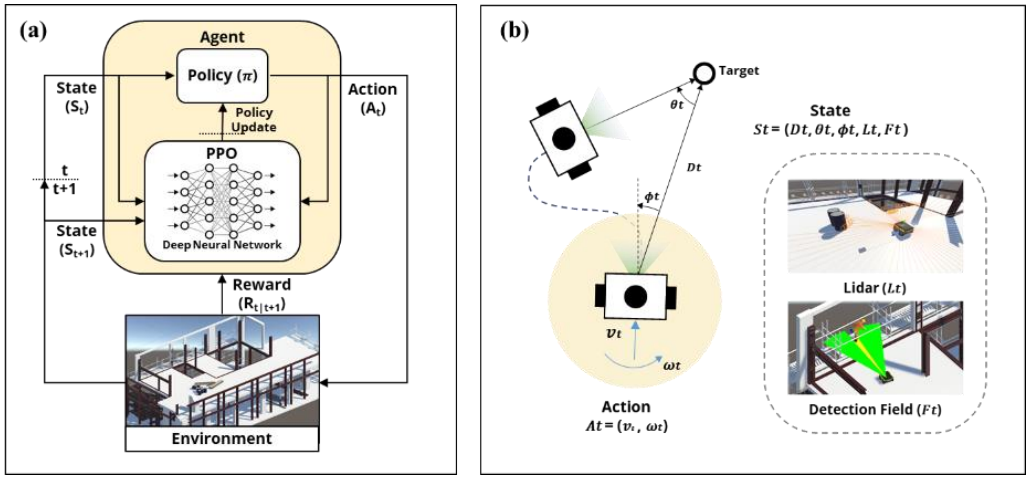


Figure 1. (a) Reinforcement learning framework, (b) Design of the state and action space.

The robot serves as an agent using an object detection model to monitor the worker's safety. During each episode, the robot, equipped with LiDAR and camera sensors, collects observational data and performs navigation and detection tasks with the primary goal of positioning the detection targets within its detection field, as illustrated in Figure 1(b). Obstacles, such as construction materials are randomly placed to hinder the robot's movement, adding complexity to the environment.

Design of State Space. The agent's state space is defined as $S_t = (D_t, \theta_t, \phi_t, L_t, F_t)$ at each time t. D_t represents the distance between the robot and the target worker, θ_t is the angle from the target worker's perspective between the robot's initial and current positions, ϕ_t is the angle between the radial direction of the observation circle and the robot's forward direction. These three components are based on the robot's spatial relationship with the target worker. L_t is the data from the lidar sensor on the robot, providing distance information about environments. F_t represents object occurrences in the robot's detection field. It detects the target or classifies other objects as obstacles. All components of the state are used for the robot's decision-making process.

Design of Action Space. The action space a_t is defined by two continuous variables, linear velocity v_t, and angular velocity w_t, enabling the robot to control its movement at each time step t. Both are normalized within the range of -1 to 1, where v_t controls forward and backward motion, while w_t governs rotation (positive for clockwise, negative for counterclockwise). This formulation ensures smooth and precise control of the robot's movement, enabling efficient navigation, obstacle avoidance, and task execution in dynamic environments.

Design of Reward. The reward system encourages the robot to navigate smoothly to its destination while avoiding collisions in an obstacle-filled environment. It also prompts the robot to continuously check for the target's presence within its detection field by adjusting its perspective. The following rewards and penalties are defined:

$$r_t(s_t, a_t) = \begin{cases} r_{detected} & if\ target\ enters\ the\ detection\ field, \\ r_{collision} & if\ collision\ occurs, \\ r_{time} \cdot \Delta t & otherwise. \end{cases}$$

The reward function $r_t(s_t, a_t)$ represents the reward obtained by the robot at time t, based on its state s_t and action a_t. This function quantitatively evaluates the outcomes of the robot's actions, guiding it to make optimal decisions during the learning process. It ensures a balance between efficient navigation, collision avoidance, and successful detection. A reward $r_{detected}$ of +10 is provided when the target successfully enters the detection field, emphasizing the value of spatial movement around the worker. The robot is encouraged to move laterally rather in a direct approach. Furthermore, to address collision avoidance, a critical aspect of autonomous robot navigation, a penalty $r_{collision}$ of -5 is applied when the robot collides with walls or obstacles, reinforcing safer navigation. Additionally, to facilitate smooth training progression, a minimal penalty r_{time} of -0.01 is introduced for time delays at every step t to discourage the robot from remaining idle in its current state and to encourage active movement.

Proximal Policy Optimization (PPO). This study employs the PPO algorithm, a reinforcement learning method well-suited for continuous control tasks. PPO uses a clipping mechanism to restrict the size of policy update steps, reducing large fluctuations and ensuring stable learning(Mammadov, 2023). It simultaneously trains actor and critic networks, where the actor focuses on decision-making and the critic evaluates actions, to improve navigation performance

and detection reliability. Additionally, PPO balances sample efficiency and stability, making it highly adaptable to dynamic environments in active monitoring tasks. These characteristics make it well-suited for active monitoring, enabling the robot to efficiently navigate around workers, adapt to environmental changes, avoid collisions, and achieve its objectives.

EXPERIMENTS

Experimental Setup. The experiment was conducted in a simulated environment to evaluate the proposed methods. Scaffolding was installed along the external walls of a structure under construction, with randomly placed construction materials serving as obstacles to increase environmental complexity and variability. Workers were positioned on the scaffolding, and the detection target was defined as the lifeline connecting their safety harnesses to the scaffolding—a critical element for verifying compliance with safety requirements. This setup was specifically designed to assess the robot's ability to detect lifelines under conditions closely resembling a real-world construction site. As shown in Figure 2-(a), four distinct scenario cases were developed to examine the position and visibility of the lifeline relative to the worker's posture. Three scenarios involved occluded lifelines, where the occlusion direction varied, occurring on the worker's left, right, or rear to verify whether the navigation policy was uniformly learned during training. Another scenario featured a fully visible lifeline in front of the worker.

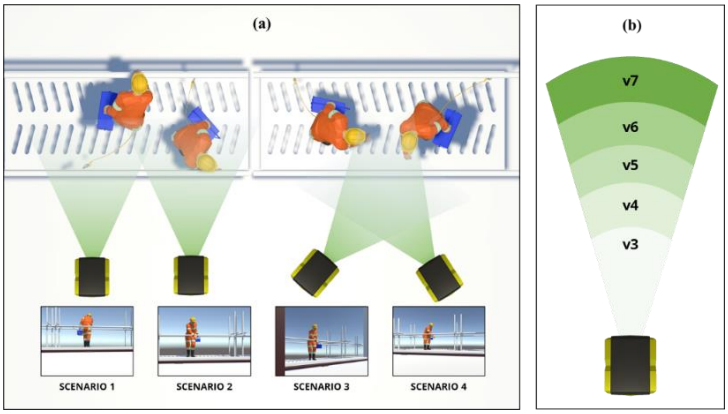


Figure 2. (a) Experiment scenario cases, (b) Robot's Detection Field.

For each scenario, experiments were conducted over 100 episodes using the Fixed-camera method, the baseline model, and the five detection field variants—v3, v4, v5, v6, and v7—shown in Figure 2-(b). At the start of each episode, a fixed camera was placed at the corner and the robot was positioned where the lifeline was not visible. The robot then navigated the environment based on each model's policy to detect the lifeline within its detection field. Detection results from the robot's final position were recorded and analyzed to compare the performance of the proposed models (v3 to v7) against the fixed camera method and the baseline across different scenarios and environmental complexities.

Evaluation Metrics. Success Rate (SR) measures the proportion of episodes in which object detection exceeds a predefined confidence threshold. It is calculated as:

$$SR = \frac{1}{N}\sum_{i=1}^{N} S_i$$

where S_i is a binary value indicating the success (1) or failure (0) of episode i, and N is the total number of episodes. The confidence threshold is set at a high confidence of 0.8. For the baseline and the proposed methods, the evaluation also included the average Trajectory Length (TL) to measure each method's navigation efficiency. Additionally, the system evaluates the minimum distance (D_{min}) to the target along the path to assess the robot's proximity to workers.

Results. The evaluation compared the fixed Camera method, a baseline model from previous research(Park et al., 2024) and multiple variations of the proposed model, each tested under different predefined detection field configurations. A summary of the results is provided in Table 1, and trajectory data for each model is presented in Figure 3.

Table 1. Experiment Result.

	w/ occlusion (n=300)			wo/occlusion (n=100)			Total Scenario		
	SR	**TL**	D_{min}	**SR**	**TL**	D_{min}	**SR**	**TL**	D_{min}
Fixed-Camera	0.04	-	13.53	0.13	-	13.53	0.07	-	13.53
Baseline	0.42	39.53	2.25	0.92	40.13	2.10	0.55	39.68	2.22
Detection Field v3	0.80	35.68	1.52	0.98	28.91	2.65	0.85	33.99	1.80
Detection Field v4	0.77	36.00	2.98	0.95	32.01	4.22	0.82	35.00	3.29
Detection Field v5	0.78	40.00	3.5	0.96	24.45	4.99	0.83	36.11	3.68
Detection Field v6	0.59	37.12	3.62	0.91	26.86	5.20	0.67	34.56	4.02
Detection Field v7	0.56	40.90	4.95	0.89	25.46	5.38	0.65	36.94	5.06

The results of this study highlight the significant impact of detection field configurations on the robot's performance. The Fixed Camera method showed a low SR of 0.07 on average and a long distance of 13.53. These results stem from the inherent characteristics of fixed cameras, which maintain a static perspective and field of view from a position far from the workers. On the other hand, the detection field v3 model demonstrated substantial improvements over the baseline in terms of success rate (SR) and trajectory length (TL). Under occlusion conditions, the v3 model achieved an SR of 0.80 and a TL of 35.68, significantly outperforming the baseline's SR of 0.42 and TL of 39.53. In disocclusion conditions, the v3 model further excelled, achieving an SR of 0.98 and a TL of 28.91, compared to the baseline's SR of 0.92 and TL of 40.13.

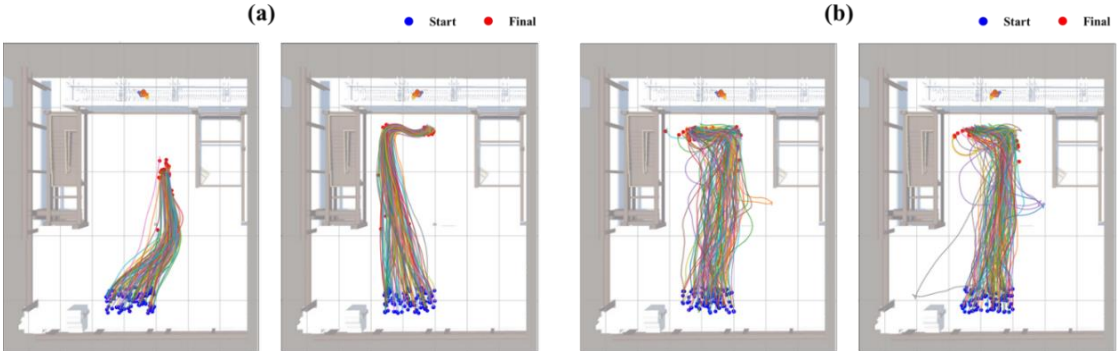


Figure 3. Robot Trajectory (a) Detection Field Method, (b) Baseline Method.

However, the v3 model's performance raised safety concerns due to its proximity to workers. The minimum distances were 1.52 meters under occlusion and 2.65 meters under disocclusion, which were significantly closer than the baseline distances of 2.25 meters and 2.10

meters, respectively. To address these concerns, the detection field was progressively extended across models v4 to v7. Kruskal-Wallis tests and Dunn's post-hoc analysis revealed significant differences among groups in terms of SR and D_{min}, although no significant differences were observed for TL. As the detection field expanded, the models improved their ability to detect targets at greater distances but experienced a decline in SR and an increase in D_{min}. Notably, the v5, v6, and v7 models maintained D_{min} values of 3.68, 4.02, and 5.06 meters, respectively, ensuring sufficient public space around workers and enabling independent movement. However, the SR of the v6 and v7 models declined significantly, dropping to 0.67 and 0.65, respectively. This highlights the trade-off between maintaining worker safety by increasing distance and ensuring high detection reliability.

DISCUSSION

This study highlights the effectiveness of an active monitoring navigation system for detecting objects under occlusion scenarios on construction sites. Unlike fixed cameras, which have limited coverage and occlusion issues, the proposed method optimizes its viewpoint autonomously without additional sensors. According to the results, the detection field strongly influences navigation, with the robot maintaining distance by moving along an arc. Extending the detection field allows the robot to monitor workers at a safe and non-intrusive distance, enhancing both psychological and physical comfort. However, excessive distance reduces detection reliability due to reliance on long-range detection. Specifically, for objects that require detailed detection, such as the lifeline used in this study, there is a threshold beyond which extending the detection range significantly reduces success rates. This suggests adjusting the detection range based on task requirements and the proximity needed for monitoring specific workers. This trade-off highlights the importance of optimizing the detection field configuration to balance safety, reliability, and navigation efficiency while considering the monitoring environment and target characteristics.

As shown in Figure 3, the significant reduction in trajectory length demonstrates that, unlike the baseline model, which maintains a fixed angular condition regardless of the target's state, detection field models dynamically adjust paths based on real-time visibility. This dynamic adjustment improves navigation efficiency by minimizing unnecessary movements. Furthermore, detection field models continuously evaluate visibility, allowing them to move toward more effective viewpoints, resulting in enhanced detection reliability under both occlusion and disocclusion conditions. These findings emphasize the role of detection field configurations in improving both detection reliability and navigation efficiency, making active monitoring navigation systems more effective across varying visibility conditions.

Comparing occlusion and disocclusion scenarios highlights the benefits of reward-based learning with a detection field. Occlusion requires closer navigation, unlike disocclusion, where targets are naturally visible. This demonstrates the rigidity of penalty-based learning, whereas reward-based learning dynamically adapts navigation and detection strategies, optimizing proximity and performance. This adaptability enhances the reliability and effectiveness of safety monitoring under varying visibility conditions.

CONCLUSION

This study presents a novel active monitoring navigation system that resolves the limitations of traditional fixed-viewpoint safety monitoring methods on construction sites. By incorporating a detection field and DRL-based navigation, the proposed system significantly enhances object

detection, particularly under occlusion. Detection field configurations allow robots to reposition dynamically, effectively overcoming visibility challenges caused by obstacles and ensuring more efficient monitoring with reliable object detection. However, there is a trade-off between detection reliability and worker safety. While closer proximity improves detection, it may raise psychological and physical safety concerns. Conversely, increasing the detection range enhances comfort but lowers success rates under occlusion. Thus, detection field configurations should be optimized based on monitoring needs and environmental complexity. As this study focuses on distance as the key factor affecting comfort, future research should examine additional human perception factors through real-world evaluations using physiological and behavioral metrics.

Adaptability is crucial in active monitoring systems. Reward-based learning mechanisms enable robots to dynamically adjust their detection and navigation strategies, improving flexibility across varying scenarios. These capabilities enhance active monitoring and support safer worker-robot interactions on construction sites.

Future research aims to extend the system by transitioning from simulation to real-world deployment. This study was conducted in a structured simulation environment, which may not fully capture real-world challenges. Factors such as sensor noise, dynamic lighting, uneven terrain, multi-worker situations, and unpredictable worker movements could impact detection reliability and navigation efficiency. Additionally, real-world occlusions are often multi-layered and dynamic, necessitating a more advanced occlusion modeling approach. To address these challenges, future work will involve testing the system under diverse environmental conditions, integrating real-world construction constraints, and evaluating its scalability for broader deployment in safety-critical workplaces. Furthermore, enhancements to detection strategies and reinforcement learning algorithms will be explored to improve adaptability to dynamic environmental factors. By expanding its scope and applicability, the proposed system has the potential to significantly advance worker safety monitoring technologies in the construction industry.

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